International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2023-0159
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First Published Online: July 18, 2023

Human Kinetics
ORIGINAL INVESTIGATION

Mental Fatigue Impairs Tackling Technique in Amateur Rugby Union Players

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Purpose: To test the effects of mental fatigue (MF) on tackling technique on the dominant and nondominant shoulders in rugby union. ***Methods:*** Twenty male amateur rugby union players and a total of 953 tackles were analyzed. A randomized crossover counterbalanced design was used across a non-MF (control) and an MF condition. During each condition, each player performed 24 tackles, divided into 4 sets of 6 tackles (3 tackles on each shoulder). In the MF condition, players performed the Stroop Task between each set of tackles. A video recording of each tackle was used to evaluate each player's technical proficiency. A score of 1 point was awarded if a specific technique was performed correctly, and 0 point was given if not. The total score, measured in arbitrary units (AU) out of 11, represents the player's overall tackling proficiency. ***Results:*** Overall, players displayed a significantly lower technical proficiency score in the MF condition compared to control (set 2: control 7.30 [7.04–7.57] AU vs MF 6.91 [6.70–7.12] AU, $P = .009$, effect size (ES) = 0.30 *small* and set 3: control 7.34 [7.11–7.57] AU vs MF 6.88 [6.66–7.11] AU, $P = .002$, ES = 0.37 *small*). For the nondominant shoulder, players had a significantly lower technical proficiency score during the MF condition at set 2 (control 7.05 [6.68–7.41] AU vs MF 6.69 [6.42–6.96] AU, $P = .047$, ES = 0.29 *small*) and set 3 (control 7.14 [6.83–7.45] AU vs MF 6.61 [6.35–6.87] AU, $P = .007$, ES = 0.49 *small*). ***Conclusions:*** MF can diminish a player's overall tackling proficiency, especially when tackling on the nondominant shoulder. The physiological mechanism for this finding may be impaired executive function and suboptimal functioning of neural signals and pathways, which result in less skillful coordination of movement. To further understand and explain MF-induced physiological changes in tackling, the feasibility of monitoring brain activity (such as electroencephalogram) and neuromuscular function (such as electromyogram) needs to be investigated. The findings from this study may also contribute to the development of more effective tackle training programs for injury prevention and performance.

Keywords: injury prevention, technical, training, skill, tackle

The tackle is a physically and psychologically challenging contest of skill between 2 or more opposing players in collision-based sports such as rugby union, rugby league, rugby sevens, and American football.[1–3]The tackle contest typically starts when a defending player(s), known as the tackler(s), attempts to impede the attacker's (the ball-carrier) progression toward the goal line and regain possession of the ball—this action is called tackling. As the main form of defense, the tackle is the most frequently occurring contact event in all the rugby football codes. In professional rugby union, for example, an average of 171 tackles occur during a match,[4] with each player involved in approximately 10 to 15 tackles per match.[2] In comparison, in community rugby, 141 tackles occur during a match.[5] Success in tackle-collision sports is therefore dependent, in part, on players' and teams' ability to repeatedly "win" the tackle contest.[6] At the same time, the nature of the tackle exposes players to high risk of injury—specifically head injuries and concussions.[2] In community rugby union, for example, the tackle event has been associated with 64% of all head injuries and 74% of all concussions.[7] In professional rugby, 53% of all concussions are associated with the tackle.[8] Tackle injuries may also be shoulder dominance dependent, with shoulder injuries more likely to occur to the nondominant shoulder than to the dominant shoulder.[9] Considering the risk of injury in the tackle, and the positive performance outcomes when players can repeatedly "win" the contest, developing effective strategies to improve player safety and performance in the tackle has become the top priority for rugby stakeholders and governing bodies.

Technical proficiency during the tackle, whether as a ball-carrier or tackler, has been shown to reduce the risk of injury and increase the likelihood of winning the contest.[10–12] Tackle injuries are also reported to occur more frequently in the second half of matches, particularly in the fourth quarter.[13] The increase in tackle injury risk during the latter stages of a match is typically credited to a decrease in tackle technique, as a consequence of increasing fatigue levels. The actual effects of physical fatigue on tackling technique have been studied in both rugby union and rugby

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league.[14,15] For example, using a simulated tackling task, Davidow et al[15] showed how the effects of physical fatigue are dependent on shoulder dominance, where tackling technique progressively declined in the nondominant shoulder when undergoing a physically fatiguing protocol.[15] Interestingly, in the nonfatigued (control) state, the players' tackling technique scores showed a progressive improvement on their nondominant shoulder. The physiological mechanism that can explain these findings is that the regulation of sites associated with tackling on the nondominant side, such as neuromuscular junctions, is not as well developed as it is on the dominant side. Thus, when changes occur in the central and peripheral systems due to acute fatigue, the sites along the psychomotor pathways are unable to adapt quickly enough to maintain proper technique.[15] It is well recognized, however, that fatigue is complex and multifaceted[16–18]; therefore, reductions in tackling technique cannot be explained by physical fatigue alone.

Mental fatigue (MF) is a psychobiological state induced by prolonged periods of demanding cognitive activity.[18–20] MF can be characterized subjectively (eg, feelings of increased tiredness, decrease in energy, and decrease in motivation), behaviorally (eg, decrease in accuracy and/or reaction time on a task), or physiologically (eg, changes in brain activity).[18,20] For example, Brownsberger et al[21] demonstrated how MF induces greater β-band activation of the frontal lobe, which led to less power output production during self-paced exercise.[21] In sport, MF has been shown to negatively affect the physical, technical, and tactical performance of athletes.[22] Most of the studies investigating the effects of MF on technical performance have been conducted in soccer.[18,19,22–27] Based on these studies and fundamental investigations such as how MF results from a failure to maintain adequate levels of dopaminergic transmission to the striatum and the anterior cingulate cortex,[28] Smith et al[23] were able to conceptualize how MF may impair players' physical, technical, tactical, and decision-making abilities in soccer.[23] For instance, MF may negatively affect players' technical abilities by reducing response times and their capacity to prepare and plan an action.[23] While the aforementioned effects of MF are specific to soccer, the proposed mechanisms are likely relevant across many sports.

In rugby union, a recent study by Davidow et al[29] demonstrated that 1 week after receiving video-based technical feedback (the retention week) for a tackling task, players showed improved tackling technique while also reporting a decrease in MF for the same task.[29] Based on these findings, the authors concluded skill learning may have occurred and players remolded their perceptual-motor landscape with new stable attractor states for tackling.[29] It follows therefore that MF has a role to play in a physical and psychologically challenging contest of skill like the tackle. However, the direct effects of MF on tackling technique are not fully understood. Therefore, the purpose of this study was to test the effects of a mentally fatiguing protocol on player's tackling technique on both their dominant and nondominant shoulder.

Methods

Participants

Twenty male rugby union players (age = 21.4 [2.6] y, height = 173.1 [5.3] cm, body mass = 84 [16.5] kg, playing experience = 11.8 [5.1] y) from the Western Province Super League A—the top amateur club contest in the Western Cape region—participated in the study. Participants were required to have participated in the previous year's season and be injury free at the time of the testing period. All the players provided their informed consent to participate in this study. Ethical approval was obtained from the University of Cape Town Faculty of Health Sciences Research Ethics Committee (HREC: 408/2017), and the study was conducted in the spirit of the Helsinki Declaration.

Study Design

A randomized, crossover, counterbalanced design was used for this study. Testing took place at the end of the competitive season. Players were required to visit the laboratory on 2 occasions—non-MF (control) and MF condition. Conditions were randomly assigned, and the order of testing was counterbalanced. During each visit, players performed a standardized warm-up and familiarized themselves with the protocol. The protocol required each player to perform 4 sets of 6 tackles (first tackle set = baseline)—3 tackles on the dominant shoulder and 3 tackles on the nondominant shoulder in random order. Shoulder order (dominant or nondominant) was indicated using a lighting system mounted behind the tackle dummy, where a light was flashed before the dummy was propelled toward the player. Shoulder dominance of each participant was self-reported on a participant information sheet prior to testing.

After the baseline tackle set, the players in the MF condition performed a 30-minute Stroop Task (ST), with two 5-minute ST top-ups following sets 1 and 2. In contrast, the players in the control condition rested passively and were allowed to read magazines for the same duration (Figure 1). Each visit was separated by at least 72 hours. In total, 953 tackles (MF = 479, control = 474) were analyzed (video footage of 7 tackles was not recorded due to technical difficulties).

Procedures

To induce MF, players performed the computer-based incongruent ST (with red mix-up) between sets of tackling. This ST displayed incongruent information to participants by having the color of the word differ from the word presented, with words in red representing their own meaning. The digital application "Stroop Test for Research" was used to administer the test (version 1.5 Andrew Novak). Previous studies have shown the 30-minute ST sufficiently induces MF.[19,24] The initial ST was performed after the baseline tackle for 30 minutes followed by 5-minute top-ups after tackle set 1 and set 2.

Figure 1 — Flow diagram of MF protocol and non-MF protocol. MF indicates mental fatigue; PR, passive rest (reading); ST, Stroop Task.

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Tackling was conducted on the University of Cape Town Tackle Contact Sport Simulator, which has been previously validated.[15,30] The tackle contact simulator resembles the physical, mental, and technical demands of one-on-one tackling without the added risk of injury experienced during live bodily tackles.[15,29,30] In brief, the simulator's objective is to replicate a front-on tackle by launching a 38-kg dummy at players at speeds comparable to those encountered during a match, without the added risk of injury experienced during live tackling drills.[31] A lighting system was mounted behind the dummy to indicate whether the tackler should use the left or right shoulder before the tackle dummy was launched toward the player. This feature added a more reactive element to tackling the simulator. A camera was positioned 3 m away from the midpoint of the simulator. Each tackle was recorded and scored by one assessor at a later stage using a standardized list of technical criteria. The technical criteria were a formulation of skill assessments in rugby league and rugby union and tackle technique coaching guidelines, which were appraised by a panel of coaches, trainers, sport clinicians, and sport scientists.[10–12,14,15,29,32] The technical criteria outline a list of observable actions for the front-on and side-on tackle, and split the tackle into 3 phases—precontact, contact, and postcontact. For example, a precontact tackler action is "head up and forward." During the analysis of a tackle, a score of 1 point was awarded if a specific technique was performed correctly and 0 point was given if not. The total score, measured in arbitrary units (AU) out of 11, represents the player's overall tackling proficiency. [10–12,14,15,29,32] The intrareliability and interreliability of the assessor (Davidow) to analyze technical proficiency on the contact simulator were tested previously.[15,29]

Before and after each tackle set, subjective ratings were measured. Immediately following a task, players were asked to rate their perceived exertion on the category ratio scale-10 (RPE),[33] perceived challenge (RPC),[34] and level of mental effort of the task just completed (ie, referring to the past). Thereafter, players were asked to rate their present state of physical and MF (referring to the present). MF, physical fatigue, and mental effort were measured on 100-mm visual analog scales. On the scales, an anchor was placed on either end of the line with "none at all" on the lowest end and "maximal" on the other end. These visual analog rating scales have previously been used in similar studies.[15,19,24]

Statistical Analyses

Mixed-effects modeling for repeated measures was performed to compare tackling proficiency between conditions (MF vs control), sets (baseline, set 1, set 2, and set 3), and between shoulder dominance (dominant shoulder vs nondominant shoulder) with a priori alpha level of significance set at $P < .05$. The mixed-effects modeling also tested the interaction between conditions and sets. Rating of exertion, RPC, reported MF, physical fatigue, and mental effort were also compared between conditions and sets using mixed-effects modeling. Cohen effect size statistic (ES) was used to determine the magnitude of the differences between conditions and sets. ESs of <0.20, 0.21 to 0.59, 0.60 to 1.19, 1.20 to 1.99, and >2.00 were considered *trivial, small, moderate, large,* and *very large*, respectively.[35] For technical proficiency scores, data are reported separately for dominant and nondominant shoulders, as well as for both shoulders combined. In addition, technical proficiency scores for each tackle phase are reported (ie, precontact, contact, and postcontact). Data are reported as estimated marginal means ± 95% CI and presented in AU. All statistical analyses were computed using STATA (version 13, StataCorp).

Results

Differences in technical proficiency scores between conditions for each shoulder and combined shoulders are shown in Figure 2. During baseline testing for combined shoulders (Figure 2A), players had a significantly higher technical proficiency score in the MF condition compared to the control condition (control 7.08 [6.82–7.35] AU vs MF 7.47 [7.21–7.72] AU, $P = .009$, ES = 0.26 *small*). After the MF protocol, this was reversed, with players displaying a significantly lower technical proficiency score in the MF condition compared to the control condition (set 2: control 7.30 [7.04–7.57] AU vs MF 6.91 [6.70–7.12] AU, $P = .009$, ES = 0.30 *small*; set 3: control 7.34 [7.11–7.57] AU vs MF 6.88 [6.66–7.11] AU, $P = .002$, ES = 0.37 *small*). In the MF condition, tackle proficiency in each set declined significantly from baseline (baseline 7.47 [7.21–7.72] AU vs set one 7.08 [6.83–7.32] AU, $P = .001$, ES = 0.35 *small*, baseline vs set two 6.91 [6.70–7.12] AU, $P < .001$, ES = 0.36 *small* and baseline vs set three 6.88 [6.66–7.11] AU, $P < .001$, ES = 0.44 *small*). Differences between conditions for

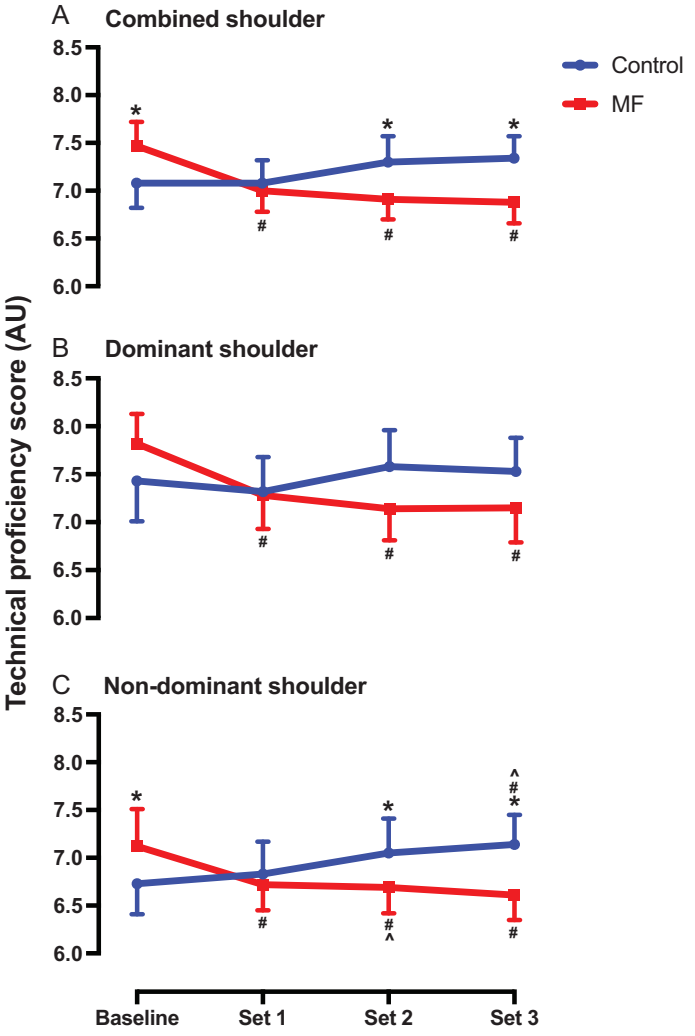

Figure 2 — Technical proficiency scores during MF and non-MF (control) conditions for both shoulders combined, dominant shoulder, and nondominant shoulder tackles. Data are reported as estimated marginal means ± 95% CI and presented in AU. AU indicates arbitrary units; MF, mental fatigue. *Different between conditions. #Different from baseline. ^Different between shoulders.

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each technical component are shown in Supplementary Tables S1–S3 (available online).

For the dominant shoulder (Figure 2B), no differences in technical proficiency between conditions were observed. Within MF condition though, tackling proficiency declined from baseline to set 1 (baseline 7.82 [7.50–8.13] AU vs set one 7.28 [6.93–7.63] AU, $P = .009$, ES = 0.41 *small*), set 2 (baseline vs set two 7.14 [6.81–7.46] AU, $P = .001$, ES = 0.55 *small*), and set 3 (baseline vs set three 7.15 [6.79–7.51] AU, $P = .001$, ES = 0.5 *small*).

For the nondominant shoulder (Figure 2C), players had a significantly lower technical proficiency score during the MF condition at set 2 (control 7.05 [6.68–7.41] AU vs MF 6.69 [6.42–6.96] AU, $P = .047$, ES = 0.29 *small*) and set 3 (control 7.14 [6.83–7.45] AU vs MF 6.61 [6.35–6.87] AU, $P = .007$, ES = 0.49 *small*). Also, during the control condition, tackling proficiency scores improved from baseline to set 3 (baseline 6.73 [6.41–7.05] AU vs set three 7.14 [6.83–.45] AU, $P = .048$, ES = 0.48 *small*) while technical proficiency scores significantly declined in the MF condition from baseline to set 1 (baseline 7.12 [6.72–7.51] AU vs set one 6.72 [6.45–6.99] AU, $P = .035$, ES = 0.31 *small*), set 2 (baseline vs set two 6.69 [6.42–6.96] AU, $P = .024$, ES = 0.33 *small*), and set 3 (baseline vs set three 6.61 [6.35–6.87] AU, $P = .006$, ES = 0.40 *small*).

Within the MF condition, the dominant shoulder scored significantly higher at set 2 (nondominant shoulder 6.69 [6.42–6.96] vs dominant shoulder 7.14 [6.81–7.556] AU, $P = .047$, ES = 0.39 *small*). Within the control condition, nondominant shoulder tackles scored significantly lower than the dominant shoulder tackles at set 3 (nondominant shoulder 7.14 [6.83–7.45] vs dominant shoulder 7.53 [7.19–7.88] AU, $P = .009$, ES = 0.32 *small*).

Differences between conditions for players' rating of exertion, technical challenge (RPC), and mental effort are shown in Figure 3. Differences between conditions for players' ratings of MF and physical fatigue are shown in Figure 4.

Discussion

This study tested the effects of a mentally fatiguing protocol on player's tackling technique on both their dominant and nondominant shoulder. We found that a state of MF can impair a player's overall tackling proficiency, and these impairments are greater when tackling on the nondominant shoulder. For the dominant shoulder, although tackling proficiency progressively declined from baseline during the MF condition, no significant differences between conditions were found. In contrast, for the nondominant shoulder, tackling proficiency progressively declined from baseline during the MF condition and was significantly lower than the control condition at set 2 and set 3. Within the MF condition, the nondominant shoulder also scored notably lower than the dominant shoulder at set 2. Previous biomechanical studies have shown weaker performances and movement patterns when tackling on the nondominant shoulder.[36,37] For example, Usman et al[36] and Seminati et al[37] found that tackling with the dominant shoulder produced higher (6% and 13%, respectively) impact forces than the nondominant shoulder. Also, when tackling with the nondominant shoulder, players seem to have poorer shoulder positional sense,[38] adopt a more passive biomechanical strategy to generate the drive needed to impede the ball-carrier, and are characterized by less head movement control, which is more flexed and laterally bent when compared to tackling on the dominant shoulder.[36] As mentioned, shoulder injuries in rugby union are also more likely to occur to the nondominant shoulder than to the dominant shoulder.[9] Differences in limb control dynamics are largely attributed to differences in the patterns of neural activation of muscles, with nondominant limbs displaying less skillful coordination of movement in terms of timing, magnitude, and direction.[39] MF, which negatively impacts executive function by activating the anterior cingulate cortex, most likely further diminishes neural signals and coordination of the nondominant side.[23]

The technical component that was significantly impaired by the MF protocol, irrespective of shoulder dominance, was "explosiveness (rapid movement) on contact." This specific feature of tackler technique has been associated with injury and poor tackle performance.[11,12,40] The inability to move rapidly upon contact after inducing MF further suggests the suboptimal functioning of the neural signals and pathway to meet the demands of efficiently performing a tackle. In soccer, poor technical performances are proposed to be attributed to negative changes in executive function, such a decrease in response time/accuracy, attention on task, action preparation/planning, and performance monitoring/adjustment.[23] It is conceivable that these MF-induced negative changes in executive function also play a role in the poor execution of rugby tackling performance.

The mentally fatiguing protocol also enhanced players' perception of physical exertion, technical challenge, and how mentally effortful the tackling task was. Players also reported both higher MF ratings and physical fatigue ratings during the MF condition. While our understanding of the mechanisms behind MF-induced impairments in sport skills is still emerging, in endurance sports, the interaction between MF, physical exertion, and mental effort is better understood. MF impacts endurance performance if it requires mental effort[41] and increases perceived exertion through 3 possible mechanisms—(1) an increase in the intensity of afferent feedback from peripheral physiological systems (the afferent feedback model), (2) increasing the intensity of central motor command (ie, motor-related cortical activity) and thus its corollary discharges (the corollary discharge model), or (3) altering the processing of these neural signals in the brain irrespective of whether they originate from the periphery or from corollary discharges of the central motor command (the combined model).[20,22] The combined model may be the most plausible physiological mechanism of the three[20] and may also explain the physiology underlying impaired tackling proficiency during the MF condition, along with corresponding increases in ratings of perceived exertion, technical challenge, and mental effort.

The purpose of the present study should be considered "use-inspired basic research."[42] That is, we sought to understand the distinct role of MF as one facet of fatigue when performing a complex and dynamic movement such as tackling. This type of research is a necessary step to develop more effective tackle training intervention studies and training models to improve practice. With that said, similar to previous work that has studied the effects of MF on sport-specific tasks,[18,22,23] the controlled nature of the study design can be considered a study limitation. For instance, to balance the risk of injury with the physical, technical, and mental demands of tackling, the tackle contact simulator only represents moderate ecological validity. The ecological validity of inducing MF using the ST for sport-specific tasks has also been questioned in the literature.[18,23] The sample of players were also high-level senior amateur club rugby union players. While tackling technique has been shown to improve with increasing playing levels (eg, senior amateur vs academy[43]), it is unclear at this stage the magnitude of the effects of MF on tackling technique at other levels (junior and elite) and for female rugby players. Future studies should also capture

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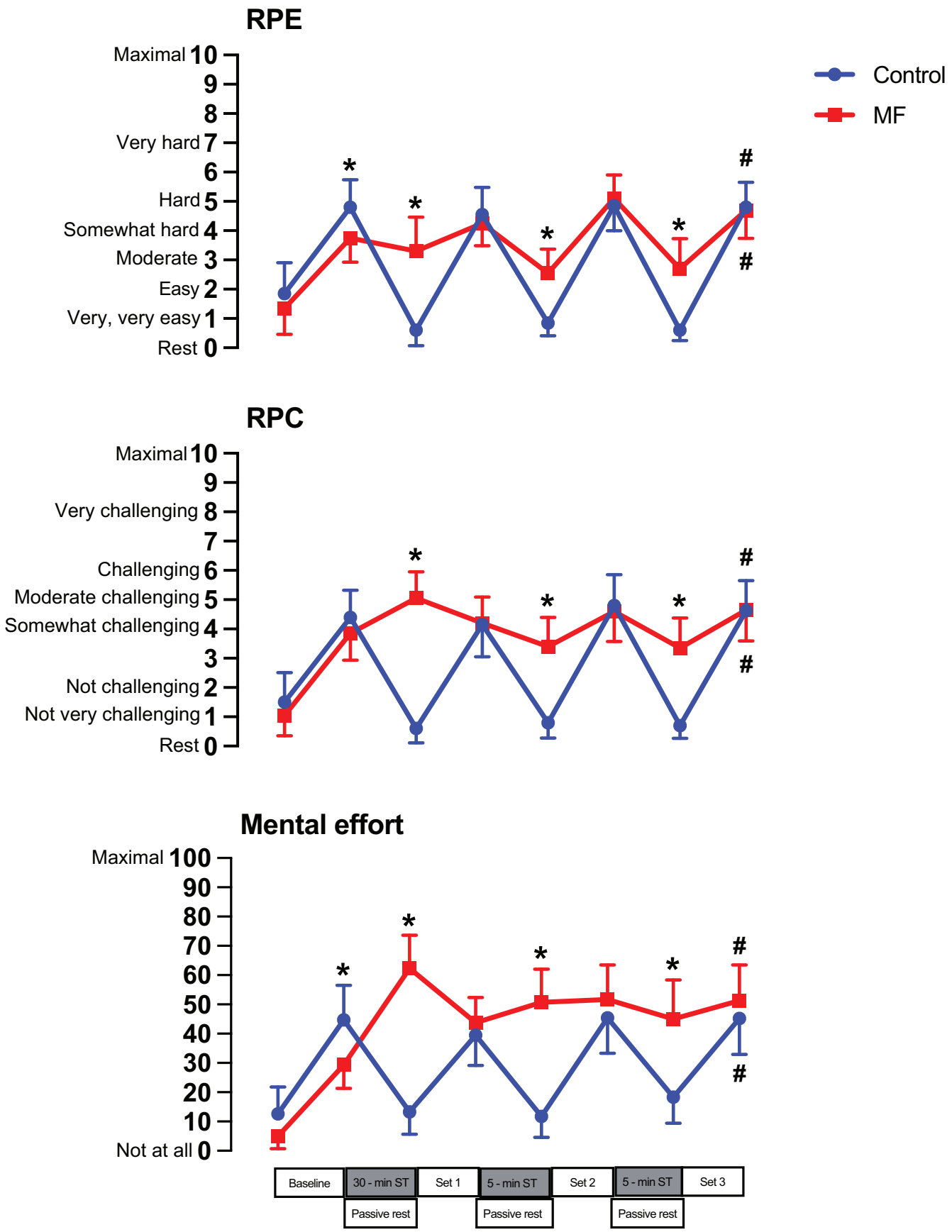

Figure 3 — RPE, RPC, and mental effort before and after each tackle set. Data are reported as estimated marginal means ± 95% CI and presented in arbitrary units. RPC indicates rating of perceived challenge; RPE, rating of perceived exertion; ST, Stroop Task. *Different between conditions ($P < .05$). #Different from baseline at set 3 ($P < .05$).

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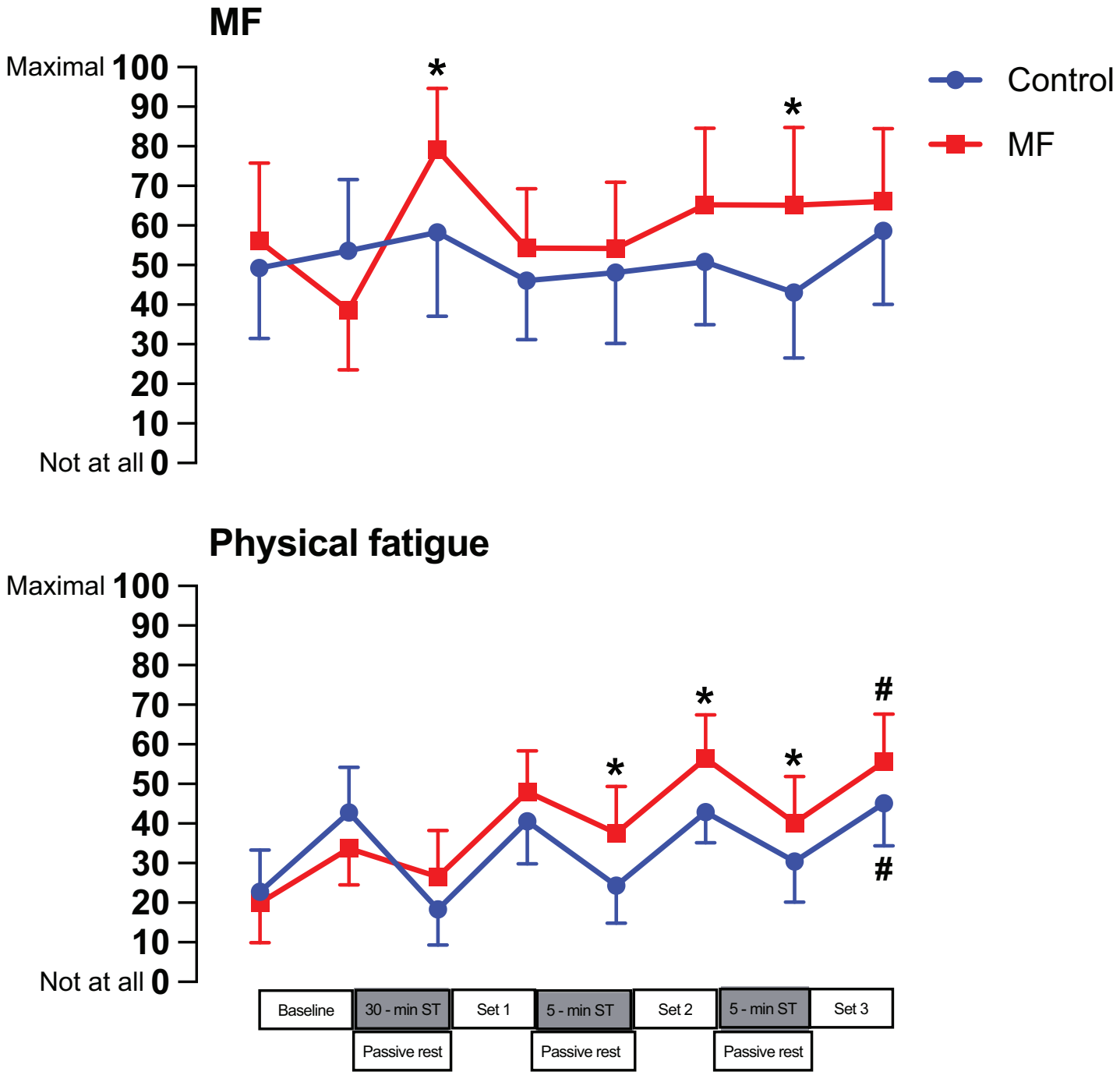

Figure 4 — MF and physical fatigue ratings before and after each tackle set. Data are reported as estimated marginal means ± 95% CI and presented in arbitrary units. MF indicates mental fatigue; ST, Stroop Task. *Different between conditions ($P < .05$). #Different from baseline at set 3 ($P < .05$).

player's tackle injury history at the time of testing, as this may influence their tackle ability. Finally, to further understand and explain the MF-induced physiological changes on tackling, the feasibility of conducting brain activity (such as electroencephalogram) and neuromuscular function (such as electromyogram) studies need to be investigated.

To reduce player's risk of injury and enhance their performance in matches, players need to develop technique and skill capacities in training.[3,44] To do this, it is recommended that players practice tackling in a state of fatigue. The fatigued state in these recommendations however is mainly focused on the physical fatigue aspect. For example, in a 5-week tackle contact training program, Hendricks et al[3] prescribe a physical conditioning block between tackle sets to develop technical capacity. To fully develop tackle capacity, based on the current findings, we recommend a MF component also be included into tackle training programs. Considering the likely interplay between mental and physical fatigue in matches and on-field training, the MF component may be a tackle-based fundamental movement task that requires motor coordination, sustained attention, cognitive processing, and perceptual

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skills.[27] For example, a set of agility ladder drills that require different foot placements/configurations in response to an external cue.[27] Another MF-inducing training strategy may be to set up a drill where players perform repeated tackling while also engaged in a highly cognitive demanding game-specific (or nongame-specific) task such as recalling tactical play configurations following verbal and/or visual cues.[45] That said, the effects of these more ecologically valid sport-specific on-field drills to induce MF compared to the commonly used ST, and how it interacts with physical fatigue, still requires further investigation.

Conclusions

Mental fatigue (MF) can impair a player's overall tackling proficiency, and these impairments are particularly prominent when tackling on the nondominant shoulder. MF also enhances players' perception of the physical, technical, and mental demands of tackling. For both dominant and nondominant shoulders, the technical component that was significantly impaired by the MF protocol was "explosiveness (rapid movement) on contact." This study can be considered "use-inspired basic research" as it aimed to improve our understanding of the distinct role of MF when performing the tackle. Considering the distinct role of MF when tackling, we recommend that an MF component be included in tackle-training programs to fully develop players' technique and skill capacities.

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